

: PROJECT TITLE :

PLACEMENT PREPARATION APP

(By SEMESTER – IX of 5th Year M.Sc.(CA & IT) 2025-26)

: Submitted by :

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1. Project Overview

- The Placement Preparation App is an Android-based mobile application designed to help students efficiently prepare for campus placements. The app provides essential resources such as aptitude questions, interview questions, coding challenges—all in one platform.
It aims to simplify the preparation process by offering a structured, easy-to-use interface where users can practice and track their progress anytime, anywhere.

2. Objectives

- To provide a centralized platform for aptitude, reasoning, verbal, and technical preparation.
- To enhance students' interview readiness through HR and technical interview questions.
- To offer real-time practice with quizzes, MCQs, and coding problems.
- To improve learning outcomes with topic-wise categorization and progress tracking.
- To reduce preparation difficulty by providing easily accessible study material and mock tests.
- To help students gain confidence before appearing for campus placement drives.

3. Scope

- In-Scope Features
 - Aptitude questions with solutions
 - Logical reasoning and verbal ability practice
 - Technical MCQs (CS/IT subjects)
 - Technical interview questions
 - Coding challenges
 - Mock tests and quizzes
 - User-friendly UI for efficient navigation
- Future Scope :
 - Integration of AI-based personalized learning
 - Daily practice reminders/notifications
 - Company-specific preparation modules
 - Online leaderboard for competitive learning
 - User login and cloud data sync
 - Video tutorials for concepts

4. Technologies Used

- Frontend
 - Android Studio – IDE for app development
 - Java – For building Android app functionalities
 - XML – For designing UI layouts
- Backend
 - Firestore – For storing questions, user scores, and progress
 - Firebase Authentication – For user login/registration
- Other Tools
 - Git/GitHub – Version control and project collaboration
 - Canva – For logo and UI design
 - Draw.io – For UML diagram

5. Problem Statement

- Students face difficulties during placement preparation because the required learning materials are scattered across many platforms, causing confusion and inefficient study patterns. There is no single, reliable app that provides aptitude practice, interview questions, mock tests, and technical questions in an organized format.
Therefore, there is a need for a unified mobile application that offers all placement preparation resources in one user-friendly and accessible platform

6. Hardware Requirements

- **For Development**

- **Processor:** Intel i5 or higher
- **RAM:** Minimum 8 GB (16 GB recommended for smooth Android Studio performance)
- **Storage:** Minimum 20 GB free space
- **Graphics:** Integrated or dedicated graphics card supporting Android Emulator
- **Operating System:** Windows / macOS / Linux

- **For Mobile Device (End User)**

- **Android Phone:** Android OS version 6.0 (Marshmallow) or above
- **RAM:** Minimum 2 GB
- **Storage:** At least 100 MB free space for installation and data
- **Screen:** Normal or large screen size supported

7. Software Requirements

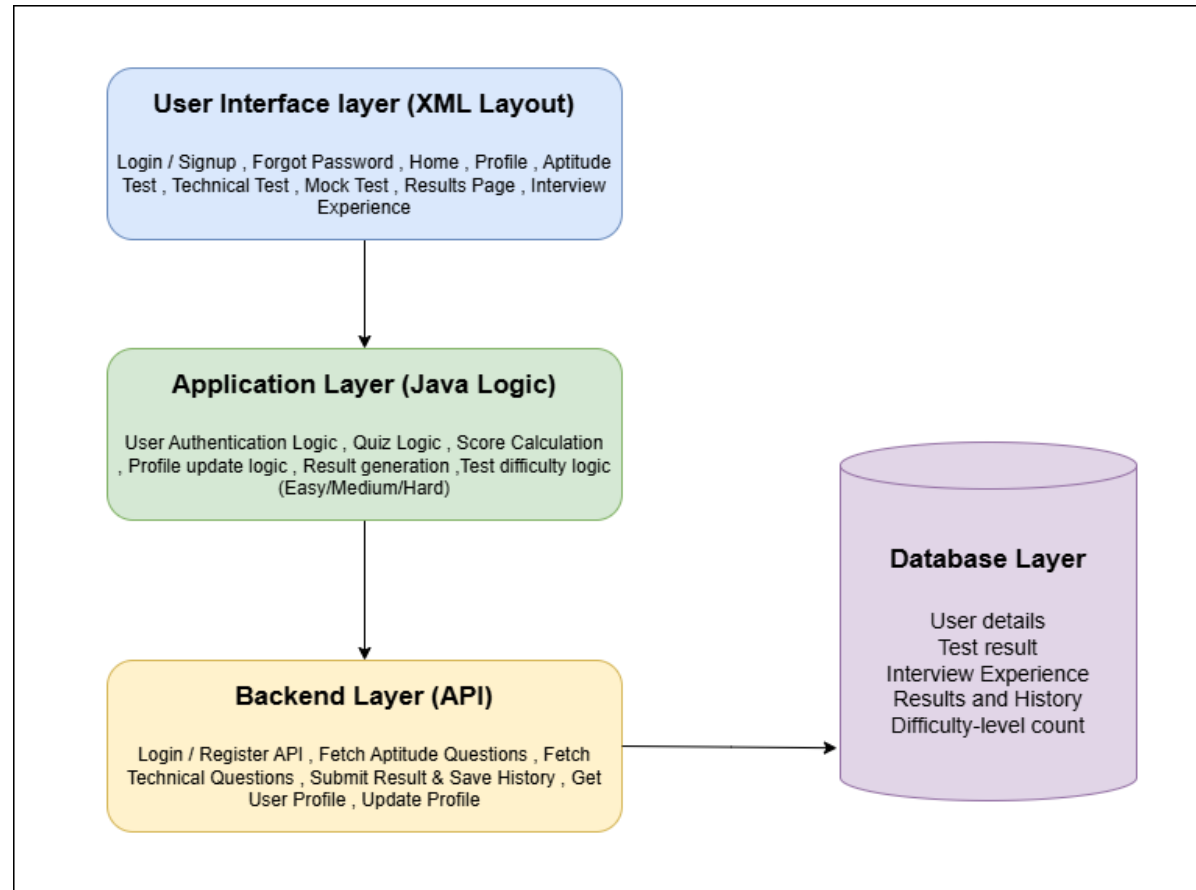
- **For Development**

- **Android Studio** (Latest Version)
- **JDK (Java Development Kit)**
- **Android SDK & Build Tools**
- **Emulator or USB Driver** for real device testing
- **Version Control:** Git/GitHub
- **Design Tools:** Canva (for UI and logo design)

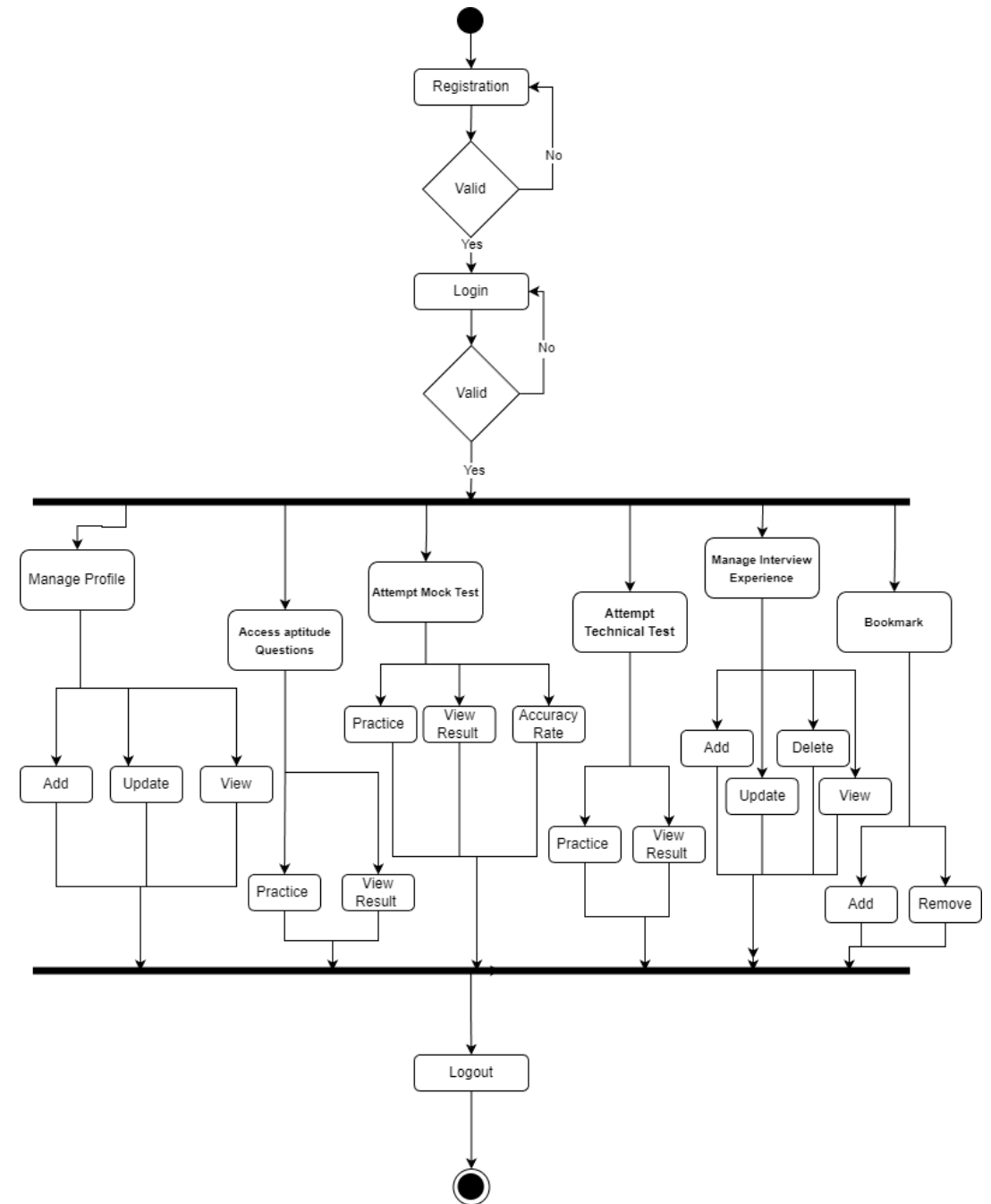
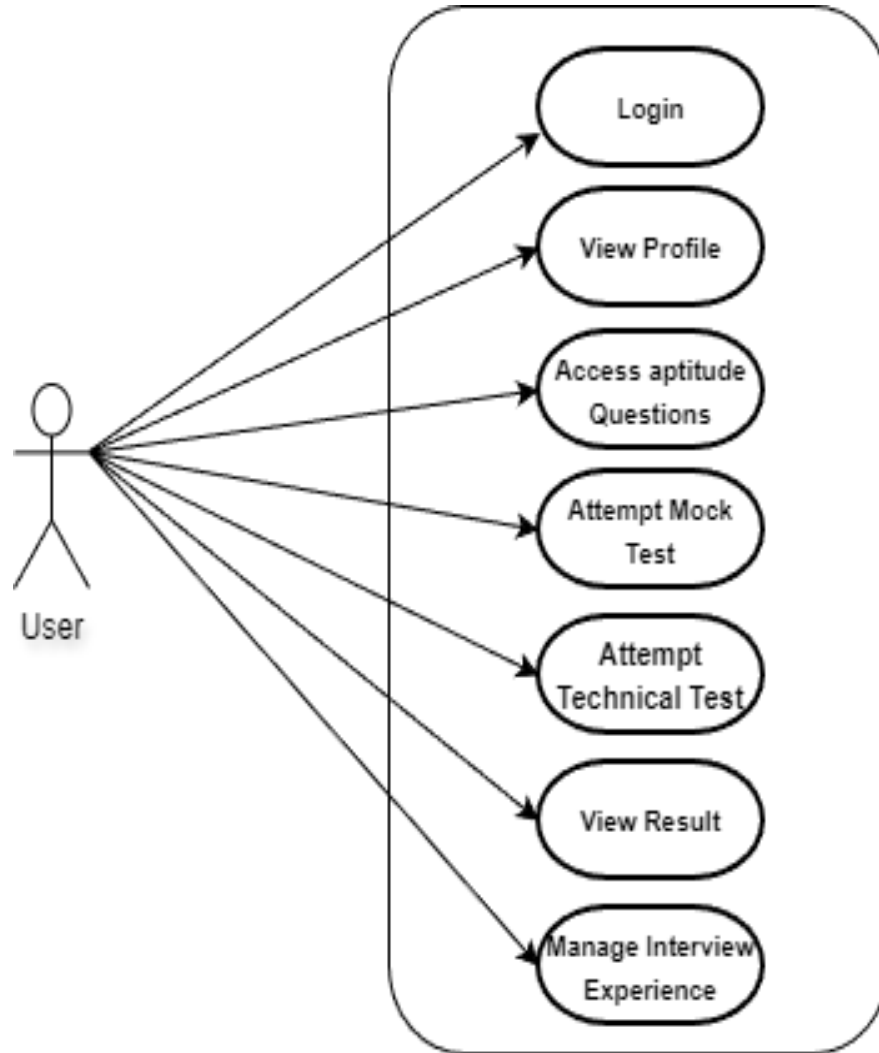
- **For Backend**

- **Firebase** (Firestore)
- **Firebase Authentication**
- **Firebase Cloud Messaging** (for notifications)
- **For Mobile Device (End User)**
- **Android OS** supporting APK
- **Internet Connection** (for loading questions, updates)

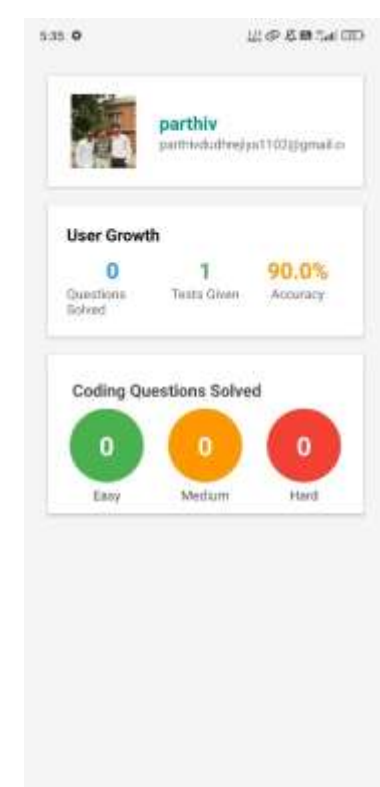
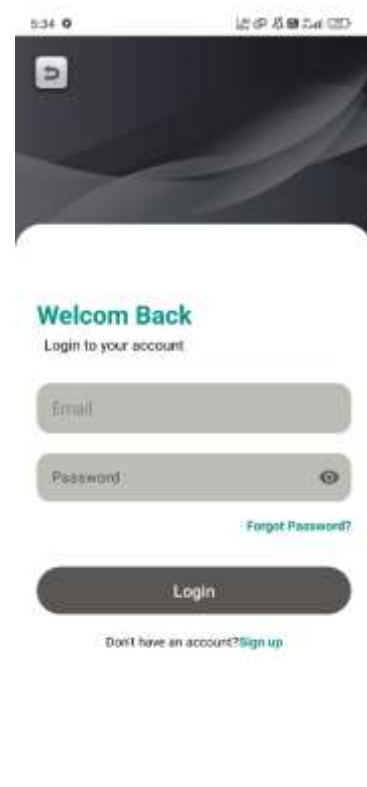
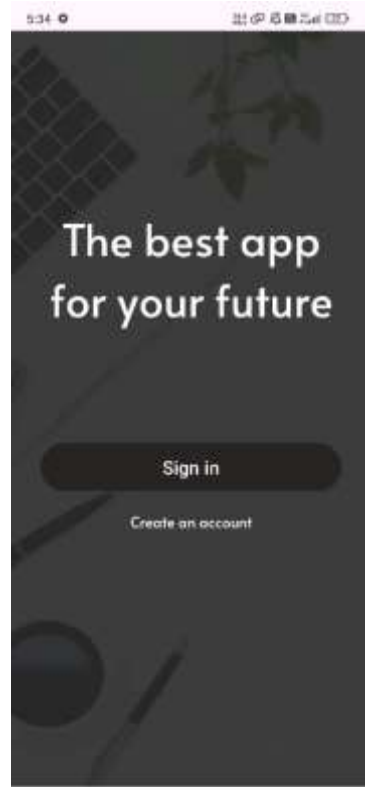
8. System Architecture

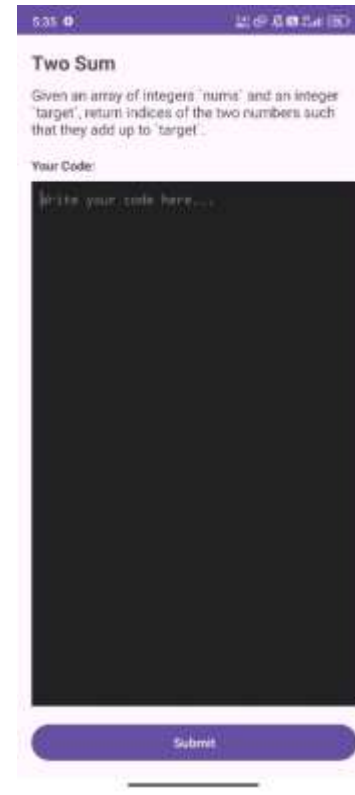
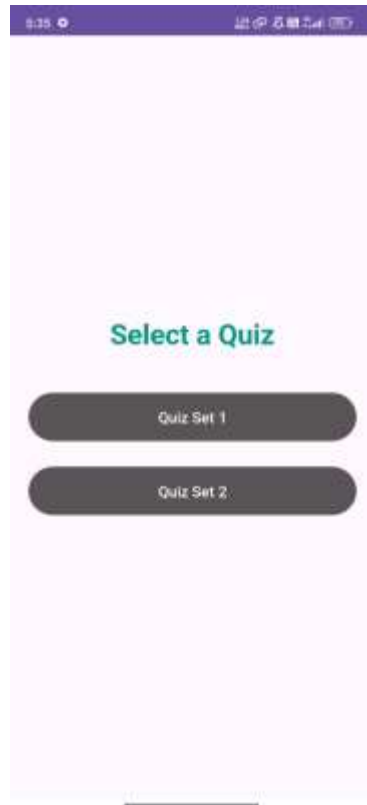


9. Use Case & Activity Diagram



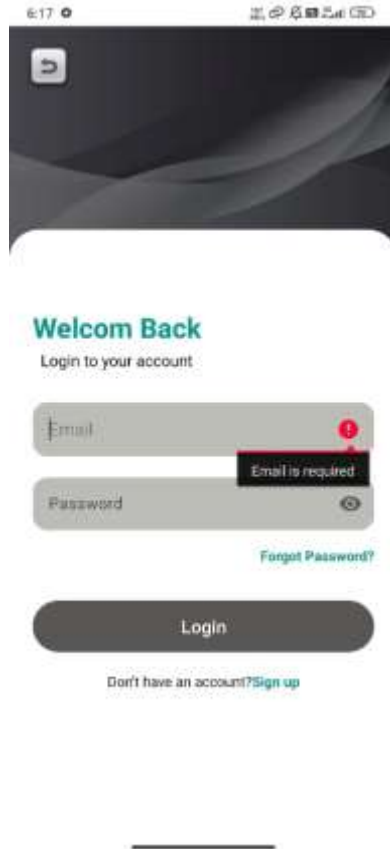
10. XML Layout Screen





11. Test Cases

1. Login : Check user login



2.Registration : Check User Registration



3.Authentication : Check user data



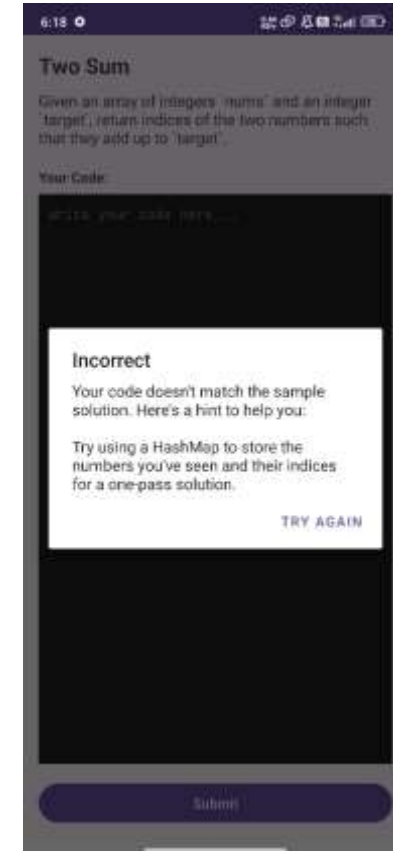
4.Password : Check Password matching



5.Question : Check correct answer



6.Technical Code : Check Code



12. Test Result

- **1. Login** : Detail is correct successfully login else get error
- **2.Registration** : detail is correct successfully register else get error
- **3.Authentication** : Credentials match successfully login
- **4.Password** : If not match give error
- **5.Question** : If correct show success else error
- **6.Technical Code** : If correct give result else show hint

13.Summary

The Placement Preparation App is designed to provide students with a centralized, user-friendly platform to prepare for campus placements efficiently. By integrating aptitude practice, reasoning questions, technical MCQs, interview preparation, and mock tests into a single mobile application, it simplifies the entire learning process. The app enhances accessibility, improves preparation quality, and supports students in developing confidence before appearing for recruitment drives. Overall, it serves as a complete preparation tool for students aiming to excel in campus placements.

14. Limitations

Although the app provides essential features for placement preparation, some limitations still exist:

- Limited offline support
- Question bank may need frequent updates to stay relevant
- Currently supports only Android devices
- No AI-based personalized recommendation system
- No real-time discussion or doubt-solving mechanism

15.Reference

- Android Studio – App development
- Firebase – Backend services
- Canva – UI & Logo design
- Draw.io - UML Diagram Tools
- Github – gist.github.com (aptitude & technical questions)